# Interval search experiments

Built 2026-10-16 16:42 PT. Experiment design by Roland Gao; runs and analysis by Claude. Code: research/cooldown_search_probe.py (runs), research/cooldown_study.py (analysis), research/make_cooldown_study.py (this PDF). Logs: logs/research_cooldown_search_probe_*, logs/research_hparam_importance_*. Batch size 2000 throughout.

## Experiment 1. Cooldown study: which 40-step lookahead ranks interval candidates like the final result?

### *Question*

An interval search picks each interval's LR by training the interval with a candidate LR, then a short lookahead (the cooldown), and scoring the result. Which cooldown ranks the candidates the way the final accuracy does? Compared, all with a 40-step lookahead (one interval at batch size 2000): no lookahead; linear decay to 0 (conv only, or all four groups); one searched constant cooldown, either for conv only with the other groups on their ground-truth schedules, or shared by all four groups; and four searched constant cooldowns (one per group).

### *Setup*

| Item | Value |
|---|---|
| Model / data | CIFAR-10 airbench, batch size 2000, 200 steps, baseline_bs2000 (conv Muon2 0.22 -> 0; head, norm_bias, whiten_bias SGD on their baseline schedules) |
| Intervals | 4 intervals of 40 steps: steps 0-40, 40-80, 80-120, 120-160 (the last interval has no room for a 40-step lookahead) |
| Candidates | 7 constant conv LRs for the interval: the 0.6^k grid point nearest the baseline's mean conv LR in that interval, and 3 grid steps either side; other groups on the baseline |
| Ground truth | Final test TTA when training continues on the baseline schedule after the interval |
| No lookahead | Test TTA right after the interval |
| Linear to 0 | Conv only: conv decays linearly from the candidate to 0 over 40 steps, the other groups continue their schedules. All groups: every group decays linearly to 0 from its current value |
| 1 searched cooldown: conv only | Conv constant at m x candidate for 40 steps; the other groups follow their baseline (ground-truth) schedules; m chosen from {1, 0.6, 0.36, 0.22, 0.13, 0.078, 0.047, 0} by the best score on that seed |
| 1 searched cooldown: shared | Constant LRs for 40 steps: conv at m x candidate, each other group at m x its mean LR over the interval; one m for all groups, chosen as above |
| 4 searched cooldowns | As above but one m per group; coordinate search from the best shared m, one group at a time over the same 8 values, until no group changes (at most 3 passes) |
| Seed | 1 training seed (0); every method uses the same prefix and data order |
| Score | Test-set TTA accuracy (%) |

### *Results*

Picked LR = the candidate with the highest score. Regret = ground truth at its best candidate minus ground truth at the picked one (pp). One seed: single-run noise is about 0.13 pp, so differences of that size between candidates are not resolved.

### *Picked LR (regret, pp)*

| Method | Interval 1 | Interval 2 | Interval 3 | Interval 4 |
|---|---|---|---|---|
| Ground truth (final TTA) | 0.078 (0.00) | 0.13 (0.00) | 0.078 (0.00) | 0.078 (0.00) |
| No lookahead | 0.078 (0.00) | 0.028 (0.34) | 0.028 (0.01) | 0.028 (0.29) |
| Linear to 0: conv only | 0.13 (0.05) | 0.078 (0.26) | 0.078 (0.00) | 0.028 (0.29) |
| Linear to 0: all four groups | 0.13 (0.05) | 0.078 (0.26) | 0.047 (0.21) | 0.028 (0.29) |
| 1 searched cooldown: conv only, others on ground truth | 0.13 (0.05) | 0.078 (0.26) | 0.047 (0.21) | 0.047 (0.21) |
| 1 searched cooldown: shared by all four groups | 0.13 (0.05) | 0.078 (0.26) | 0.047 (0.21) | 0.047 (0.21) |
| 4 searched cooldowns (one per group) | 0.13 (0.05) | 0.078 (0.26) | 0.047 (0.21) | 0.047 (0.21) |

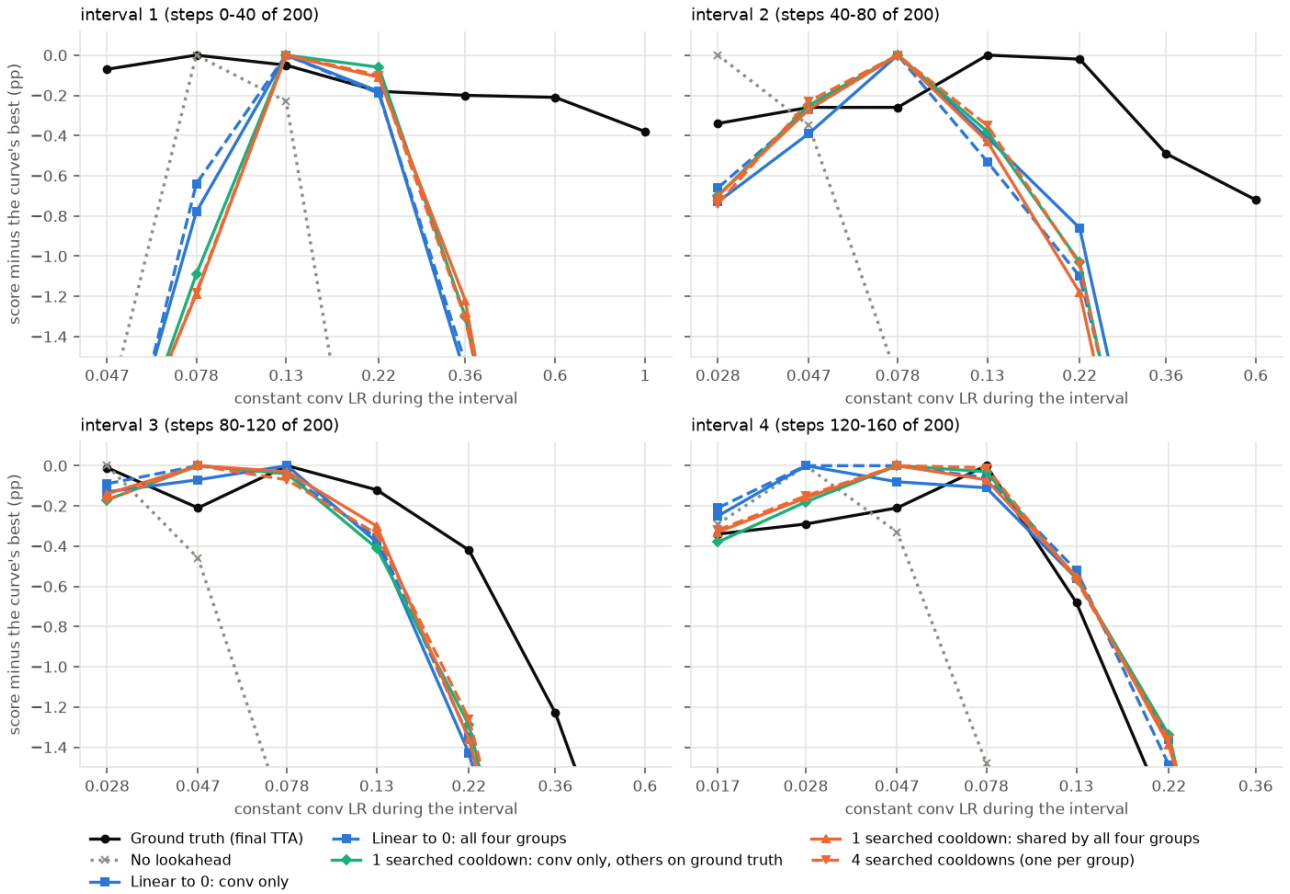



Figure 1. Each method's score per candidate, relative to its own best (the point at 0 is its pick).

### *Interval 1: score per candidate conv LR (bold: each row's best)*

| Method | 0.047 | 0.078 | 0.13 | 0.22 | 0.36 | 0.6 | 1 |
|---|---|---|---|---|---|---|---|
| Ground truth (final TTA) | 93.99 | **94.06** | 94.01 | 93.88 | 93.86 | 93.85 | 93.68 |
| No lookahead | 81.57 | **83.35** | 83.12 | 80.45 | 68.80 | 56.07 | 35.18 |
| Linear to 0: conv only | 89.04 | 90.59 | **91.37** | 91.19 | 89.78 | 87.38 | 84.39 |
| Linear to 0: all four groups | 88.85 | 90.63 | **91.27** | 91.08 | 89.74 | 87.41 | 84.10 |
| 1 searched cooldown: conv only, others on ground truth | 88.58 | 89.83 | **90.92** | 90.86 | 89.62 | 87.45 | 85.95 |
| 1 searched cooldown: shared by all four groups | 88.65 | 89.70 | **90.89** | 90.78 | 89.67 | 87.16 | 85.13 |
| 4 searched cooldowns (one per group) | 88.79 | 89.89 | **91.07** | 90.97 | 89.76 | 87.65 | 86.13 |

| Candidate | Conv only: chosen m (seeds) | Shared: chosen m (seeds) | 4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds) |
|---|---|---|---|
| 0.047 | 0.6 x1 | 0.6 x1 | (1, 1, 0.6, 0.36) x1 |
| 0.078 | 0.36 x1 | 0.36 x1 | (0.36, 1, 1, 0.22) x1 |
| 0.13 | 0.22 x1 | 0.22 x1 | (0.22, 1, 0.22, 0.22) x1 |
| 0.22 | 0.22 x1 | 0.22 x1 | (0.22, 0.36, 0.22, 1) x1 |
| 0.36 | 0.13 x1 | 0.13 x1 | (0.13, 0.047, 0.13, 0.13) x1 |
| 0.6 | 0.078 x1 | 0.078 x1 | (0.078, 1, 0.36, 0.078) x1 |
| 1 | 0.078 x1 | 0.078 x1 | (0.078, 0.6, 0.078, 0.6) x1 |

### *Interval 2: score per candidate conv LR (bold: each row's best)*

| Method | 0.028 | 0.047 | 0.078 | 0.13 | 0.22 | 0.36 | 0.6 |
|---|---|---|---|---|---|---|---|
| Ground truth (final TTA) | 93.76 | 93.84 | 93.84 | **94.10** | 94.08 | 93.61 | 93.38 |
| No lookahead | **91.12** | 90.77 | 89.51 | 87.43 | 82.45 | 71.27 | 57.08 |
| Linear to 0: conv only | 92.25 | 92.59 | **92.98** | 92.57 | 92.12 | 90.18 | 87.94 |
| Linear to 0: all four groups | 92.27 | 92.66 | **92.93** | 92.40 | 91.83 | 90.11 | 87.78 |
| 1 searched cooldown: conv only, others on ground truth | 92.16 | 92.61 | **92.86** | 92.48 | 91.83 | 89.82 | 87.93 |
| 1 searched cooldown: shared by all four groups | 92.18 | 92.61 | **92.88** | 92.45 | 91.70 | 89.73 | 87.84 |
| 4 searched cooldowns (one per group) | 92.20 | 92.71 | **92.94** | 92.59 | 91.90 | 89.87 | 87.98 |

| Candidate | Conv only: chosen m (seeds) | Shared: chosen m (seeds) | 4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds) |
|---|---|---|---|
| 0.028 | 0.6 x1 | 0.6 x1 | (0.6, 0.6, 1, 0.6) x1 |
| 0.047 | 0.36 x1 | 0.36 x1 | (0.36, 1, 0.047, 0.36) x1 |
| 0.078 | 0.22 x1 | 0.22 x1 | (0.22, 0.13, 0.22, 0.22) x1 |
| 0.13 | 0.22 x1 | 0.22 x1 | (0.22, 1, 0.36, 0) x1 |
| 0.22 | 0.13 x1 | 0.13 x1 | (0.13, 1, 1, 0.078) x1 |
| 0.36 | 0.13 x1 | 0.13 x1 | (0.13, 0.6, 0, 0.047) x1 |
| 0.6 | 0.078 x1 | 0.078 x1 | (0.078, 0.047, 0.13, 0.078) x1 |

### *Interval 3: score per candidate conv LR (bold: each row's best)*

| Method | 0.028 | 0.047 | 0.078 | 0.13 | 0.22 | 0.36 | 0.6 |
|---|---|---|---|---|---|---|---|
| Ground truth (final TTA) | 93.99 | 93.79 | **94.00** | 93.88 | 93.58 | 92.77 | 91.54 |
| No lookahead | **92.59** | 92.13 | 90.81 | 87.68 | 82.32 | 71.78 | 56.20 |
| Linear to 0: conv only | 93.26 | 93.32 | **93.39** | 93.01 | 91.96 | 90.45 | 87.63 |
| Linear to 0: all four groups | 93.28 | **93.37** | 93.33 | 93.01 | 92.02 | 90.23 | 87.59 |
| 1 searched cooldown: conv only, others on ground truth | 93.16 | **93.33** | 93.29 | 92.92 | 92.04 | 90.31 | 88.61 |
| 1 searched cooldown: shared by all four groups | 93.13 | **93.27** | 93.24 | 92.97 | 91.91 | 90.29 | 88.64 |
| 4 searched cooldowns (one per group) | 93.18 | **93.35** | 93.28 | 93.01 | 92.09 | 90.44 | 88.76 |

| Candidate | Conv only: chosen m (seeds) | Shared: chosen m (seeds) | 4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds) |
|---|---|---|---|
| 0.028 | 0.36 x1 | 0.22 x1 | (0.22, 0.13, 1, 0.22) x1 |
| 0.047 | 0.22 x1 | 0.36 x1 | (0.22, 1, 1, 0.36) x1 |
| 0.078 | 0.22 x1 | 0.22 x1 | (0.22, 1, 0.36, 0.22) x1 |
| 0.13 | 0.22 x1 | 0.22 x1 | (0.22, 1, 0.22, 0.22) x1 |
| 0.22 | 0.22 x1 | 0.13 x1 | (0.22, 0.36, 1, 0.13) x1 |
| 0.36 | 0.13 x1 | 0.13 x1 | (0.13, 0.078, 0.6, 0.13) x1 |
| 0.6 | 0.078 x1 | 0.13 x1 | (0.078, 0.36, 0.13, 0.13) x1 |

***Interval 4: score per candidate conv LR (bold: each row's best)***

| Method | 0.017 | 0.028 | 0.047 | 0.078 | 0.13 | 0.22 | 0.36 |
|---|---|---|---|---|---|---|---|
| Ground truth (final TTA) | 93.65 | 93.70 | 93.78 | **93.99** | 93.31 | 92.23 | 89.58 |
| No lookahead | 92.89 | **93.18** | 92.85 | 91.70 | 88.84 | 83.30 | 72.23 |
| Linear to 0: conv only | 93.55 | **93.80** | 93.72 | 93.69 | 93.24 | 92.44 | 90.63 |
| Linear to 0: all four groups | 93.55 | **93.76** | **93.76** | 93.70 | 93.24 | 92.27 | 90.57 |
| 1 searched cooldown: conv only, others on ground truth | 93.51 | 93.71 | **93.89** | 93.86 | 93.33 | 92.55 | 90.70 |
| 1 searched cooldown: shared by all four groups | 93.56 | 93.73 | **93.89** | 93.82 | 93.34 | 92.50 | 90.89 |
| 4 searched cooldowns (one per group) | 93.62 | 93.79 | **93.94** | 93.93 | 93.37 | 92.58 | 90.93 |

| Candidate | Conv only: chosen m (seeds) | Shared: chosen m (seeds) | 4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds) |
|---|---|---|---|
| 0.017 | 0.6 x1 | 0.6 x1 | (0.6, 1, 1, 0.6) x1 |
| 0.028 | 0.6 x1 | 0.6 x1 | (0.6, 1, 0.36, 0.6) x1 |
| 0.047 | 0.36 x1 | 0.36 x1 | (0.36, 0.22, 1, 0.36) x1 |
| 0.078 | 0.22 x1 | 0.22 x1 | (0.22, 0.047, 0.22, 0.22) x1 |
| 0.13 | 0.13 x1 | 0.13 x1 | (0.13, 0.22, 0.13, 0.13) x1 |
| 0.22 | 0.13 x1 | 0.13 x1 | (0.13, 1, 0.078, 0.13) x1 |
| 0.36 | 0.13 x1 | 0.13 x1 | (0.13, 0.078, 1, 0.13) x1 |

### *Observations*

One seed: single-run noise is about 0.13 pp, so ground-truth differences of that size are not resolved. Interval 1's ground truth is flat (93.85-94.06 from 0.047 to 0.6) and interval 3's is non-monotone (0.028: 93.99, 0.047: 93.79, 0.078: 94.00), so regrets of about 0.2 pp or less at those intervals are within noise.

All five 40-step lookahead variants pick the same LR at intervals 1 and 2, and the three searched-constant variants (conv only, shared, four per group) pick the same LR at every interval. How the other three groups cool down (continuing their ground-truth schedules, one shared multiplier, or their own multipliers) never changed the pick. Four searched multipliers score slightly higher in absolute terms than one but rank the candidates the same way.

The 40-step lookahead picks a smaller LR than the ground truth where the ground truth is clearly peaked: interval 2 (all variants pick 0.078 where 0.13 is best, 0.26 pp regret) and interval 4 (0.028-0.047 where 0.078 is best, 0.21-0.29 pp). No lookahead is worse at intervals 2 and 4 (0.34 / 0.29 pp) and always picks the smallest or second-smallest candidate after interval 1.

Linear decay and searched constants differ only at intervals 3 and 4, in opposite directions (linear conv-only is better at interval 3, searched constants at interval 4), both by about 0.2-0.3 pp: not resolved with one seed.

## Experiment 2. Hparam importance from scratch

### *Question*

How important is each of the four LRs when they are searched jointly from scratch, and does the answer depend on the main interval length?

### *Setup*

| Item | Value |
|---|---|
| Model / data | baseline_bs2000 except the four trained LRs (conv, head, norm_bias, whiten_bias), which are constant for the first L training steps from initialization |
| Main interval L | 1, 2, 5, 10, 40 steps; the score is taken right after the L steps (no lookahead) |
| Joint search | From scratch: every LR starts at 1.0. Scan each LR over x10^k, k = -3..3 (others fixed), keep the best, snap to the 0.6^k grid, then coordinate-climb on the grid (one step up / down, move while the score improves) until no LR moves: the center |
| Sweep | From the center, each LR alone over 5 grid points on each side (others at the center). Offset +1 = one grid step larger (x1/0.6), -1 = one step smaller (x0.6) |
| Seed / score | 1 seed (0); test-set TTA accuracy (%); the test cross-entropy loss is logged too |

### *Joint-search centers*

| L (steps) | Center: conv, head, norm_bias, whiten_bias | Center TTA | Evaluations |
|---|---|---|---|
| 1 | 0.13, 60, 0.00078, 60 | 25.43 | 73 |
| 2 | 0.22, 1, 1.7, 1300 | 36.89 | 71 |
| 5 | 0.078, 0.006, 2100, 1300 | 52.00 | 76 |
| 10 | 0.078, 0.017, 2100, 13 | 59.47 | 70 |
| 40 | 0.078, 2100, 280, 2100 | 84.60 | 73 |

### *Importance: TTA drop from the center (pp) at 1 step / 2 steps (mean of both sides) / worst within 5 steps*

| LR | L = 1 | L = 2 | L = 5 | L = 10 | L = 40 |
|---|---|---|---|---|---|
| conv | 0.58 / 2.01 / 16.42 | 1.00 / 2.90 / 19.69 | 3.20 / 7.96 / 32.96 | 2.02 / 4.11 / 29.37 | 1.95 / 5.26 / 49.82 |
| head | 0.03 / 0.14 / 0.29 | 0.02 / 0.02 / 0.08 | 0.10 / 0.12 / 0.18 | 0.32 / 0.49 / 0.67 | 0.54 / 1.65 / 62.56 |
| norm_bias | 0.00 / 0.00 / 0.00 | 0.05 / 0.02 / 0.08 | 0.51 / 1.46 / 28.09 | 1.12 / 2.77 / 28.84 | 0.43 / 0.84 / 6.98 |
| whiten_bias | 0.01 / 0.03 / 0.09 | 0.15 / 0.85 / 4.48 | 0.56 / 1.02 / 3.45 | 0.51 / 0.58 / 2.00 | 0.46 / 0.96 / 2.44 |

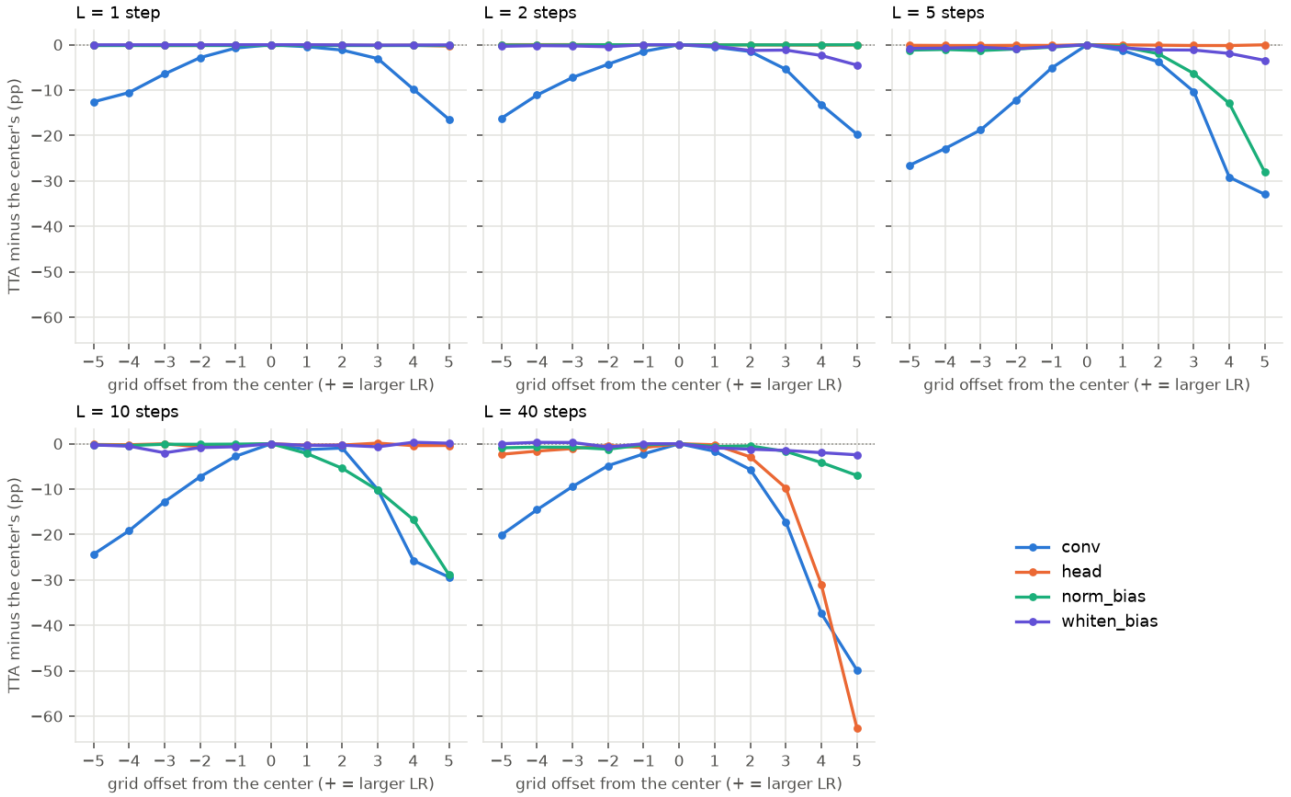



Figure 2. One-at-a-time sweeps around each L's joint-search center.

***L = 1: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 25.43)***

| LR | -5 | -4 | -3 | -2 | -1 | +0 | +1 | +2 | +3 | +4 | +5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| conv (0.13) | -12.58 | -10.53 | -6.44 | -2.84 | -0.73 | +0.00 | -0.43 | -1.18 | -3.06 | -9.80 | -16.42 |
| head (60) | -0.15 | -0.14 | -0.17 | -0.16 | -0.03 | +0.00 | -0.03 | -0.11 | -0.16 | -0.12 | -0.29 |
| norm_bias (0.00078) | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 |
| whiten_bias (60) | -0.03 | -0.02 | -0.01 | -0.01 | -0.01 | +0.00 | -0.01 | -0.05 | -0.09 | -0.08 | -0.04 |

***L = 2: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 36.89)***

| LR | -5 | -4 | -3 | -2 | -1 | +0 | +1 | +2 | +3 | +4 | +5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| conv (0.22) | -16.19 | -11.09 | -7.21 | -4.29 | -1.48 | +0.00 | -0.52 | -1.50 | -5.41 | -13.19 | -19.69 |
| head (1) | +0.01 | +0.01 | +0.02 | +0.00 | -0.01 | +0.00 | -0.03 | -0.03 | -0.05 | -0.08 | -0.04 |
| norm_bias (1.7) | -0.07 | -0.08 | -0.05 | -0.01 | -0.04 | +0.00 | -0.05 | -0.03 | -0.03 | -0.02 | -0.01 |
| whiten_bias (1300) | -0.34 | -0.20 | -0.26 | -0.44 | -0.01 | +0.00 | -0.29 | -1.26 | -1.18 | -2.37 | -4.48 |

***L = 5: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 52.00)***

| LR | -5 | -4 | -3 | -2 | -1 | +0 | +1 | +2 | +3 | +4 | +5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| conv (0.078) | -26.55 | -22.87 | -18.78 | -12.17 | -5.13 | +0.00 | -1.28 | -3.75 | -10.31 | -29.17 | -32.96 |
| head (0.006) | -0.13 | -0.17 | -0.17 | -0.14 | -0.16 | +0.00 | -0.04 | -0.10 | -0.17 | -0.18 | -0.02 |
| norm_bias (2100) | -1.19 | -1.04 | -1.29 | -0.94 | -0.54 | +0.00 | -0.47 | -1.97 | -6.33 | -12.92 | -28.09 |
| whiten_bias (1300) | -0.81 | -0.75 | -0.58 | -0.91 | -0.41 | +0.00 | -0.71 | -1.13 | -1.17 | -1.92 | -3.45 |

***L = 10: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 59.47)***

| LR | -5 | -4 | -3 | -2 | -1 | +0 | +1 | +2 | +3 | +4 | +5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| conv (0.078) | -24.35 | -19.17 | -12.77 | -7.28 | -2.76 | +0.00 | -1.28 | -0.95 | -10.13 | -25.72 | -29.37 |
| head (0.017) | -0.17 | -0.23 | -0.02 | -0.67 | -0.31 | +0.00 | -0.34 | -0.30 | +0.13 | -0.42 | -0.38 |
| norm_bias (2100) | -0.29 | -0.39 | -0.12 | -0.13 | -0.09 | +0.00 | -2.15 | -5.42 | -10.23 | -16.70 | -28.84 |
| whiten_bias (13) | -0.26 | -0.53 | -2.00 | -0.85 | -0.67 | +0.00 | -0.36 | -0.32 | -0.68 | +0.32 | +0.11 |

***L = 40: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 84.60)***

| LR | -5 | -4 | -3 | -2 | -1 | +0 | +1 | +2 | +3 | +4 | +5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| conv (0.078) | -20.08 | -14.54 | -9.42 | -4.83 | -2.24 | +0.00 | -1.67 | -5.69 | -17.22 | -37.36 | -49.82 |
| head (2100) | -2.33 | -1.63 | -1.07 | -0.40 | -0.85 | +0.00 | -0.23 | -2.90 | -9.74 | -31.04 | -62.56 |
| norm_bias (280) | -0.90 | -0.75 | -0.77 | -1.20 | -0.24 | +0.00 | -0.62 | -0.48 | -1.69 | -4.15 | -6.98 |
| whiten_bias (2100) | +0.00 | +0.30 | +0.27 | -0.72 | -0.01 | +0.00 | -0.91 | -1.21 | -1.47 | -1.97 | -2.44 |

## Experiment 3. Representation quality of schedule families

### *Question*

How close to the baseline can each family of conv LR schedules get, under no transform and under the ramp, reverse ramp and wave transforms, when searched by the same global search?

### *Setup*

| Item | Value |
|---|---|
| Model / data | baseline_bs2000; only the conv LR schedule is searched, the other groups stay on the baseline. The transform multiplies the conv LR the search specifies |
| Conditions | No transform; ramp x 4^(2x-1) (x0.25 -> x4); reverse ramp x 4^(1-2x) (x4 -> x0.25); wave x 2^sin(2 pi x) (x1 -> x2 -> x0.5 -> x1); x = fraction of training |
| Families | 5 constant intervals (5 values); linear decay to 0 (initial LR); power decay lr x (1-t)^p (lr, p; p = 1 is linear); 2 knots at t = 0, 0.5; 3 knots at t = 0, 1/3, 2/3; 5 knots at t = 0, 0.2, 0.4, 0.6, 0.8 (knot values; linear between knots, 0 at the end) |
| Search | Generalized global_neighbour (global_family): coordinate search over the family's parameters on multiplicative grids (LR values x0.6, p x0.8), one grid step at a time, moving while the score improves; every candidate is a full training run on seed 0 from the same initial state |
| Start | The oracle spec (baseline conv schedule divided by the transform) projected onto the family: interval means or knot values snapped to the grid; linear / power start at 0.22 with p = 1 |
| Evaluation | Start and best schedules retrained on held-out seeds 100-107 (mean test TTA; not part of the search) |

### *References (held-out TTA, seeds 100-107)*

| | No transform | Ramp | Reverse ramp | Wave |
|---|---|---|---|---|
| Baseline linear decay, no transform (the target) | 93.92 | 93.92 | 93.92 | 93.92 |
| Baseline spec kept under the transform (do-nothing) | 93.92 | 93.42 | 93.57 | 93.72 |

### *Held-out TTA of the best schedule (start = the projected oracle before search; search cost)*

| Family | No transform | Ramp | Reverse ramp | Wave |
|---|---|---|---|---|
| 5 constant intervals | **93.62** (start 93.52; 18 runs, 182 s) | **93.65** (start 93.32; 28 runs, 285 s) | **93.86** (start 93.81; 28 runs, 279 s) | **93.82** (start 93.11; 22 runs, 222 s) |
| Linear decay (1 param) | **93.86** (start 93.92; 3 runs, 31 s) | **93.42** (start 93.42; 3 runs, 31 s) | **93.59** (start 93.57; 4 runs, 41 s) | **93.91** (start 93.72; 3 runs, 31 s) |
| Power decay (lr, p) | **93.86** (start 93.92; 8 runs, 81 s) | **93.95** (start 93.42; 14 runs, 141 s) | **93.59** (start 93.57; 6 runs, 61 s) | **93.91** (start 93.72; 5 runs, 51 s) |
| 2 knots (t = 0, 0.5) | **93.86** (start 93.86; 5 runs, 51 s) | **93.77** (start 93.14; 9 runs, 91 s) | **93.88** (start 93.85; 8 runs, 83 s) | **93.83** (start 93.72; 12 runs, 121 s) |
| 3 knots (t = 0, 1/3, 2/3) | **93.86** (start 93.94; 12 runs, 121 s) | **93.68** (start 93.74; 10 runs, 101 s) | **93.78** (start 93.90; 13 runs, 131 s) | **94.06** (start 93.90; 7 runs, 73 s) |
| 5 knots (t = 0, 0.2, ..., 0.8) | **93.94** (start 93.97; 13 runs, 131 s) | **93.96** (start 93.91; 17 runs, 170 s) | **93.91** (start 93.91; 11 runs, 111 s) | **93.94** (start 93.97; 11 runs, 111 s) |

### *Best parameter values (LR values in order; power: lr, p)*

| Family | No transform | Ramp | Reverse ramp | Wave |
|---|---|---|---|---|
| 5 constant intervals | 0.13, 0.22, 0.078, 0.047, 0.028 | 0.36, 0.13, 0.078, 0.028, 0.0036 | 0.028, 0.078, 0.13, 0.078, 0.047 | 0.078, 0.078, 0.13, 0.078, 0.017 |
| Linear decay (1 param) | 0.13 | 0.22 | 0.078 | 0.13 |
| Power decay (lr, p) | 0.13, 0.64 | 0.36, 2 | 0.078, 1 | 0.13, 1 |
| 2 knots (t = 0, 0.5) | 0.22, 0.13 | 0.6, 0.078 | 0.078, 0.078 | 0.13, 0.078 |
| 3 knots (t = 0, 1/3, 2/3) | 0.13, 0.22, 0.047 | 0.36, 0.13, 0.047 | 0.078, 0.078, 0.047 | 0.13, 0.078, 0.13 |
| 5 knots (t = 0, 0.2, ..., 0.8) | 0.13, 0.13, 0.13, 0.078, 0.047 | 1, 0.36, 0.22, 0.047, 0.017 | 0.047, 0.078, 0.078, 0.13, 0.13 | 0.13, 0.078, 0.078, 0.13, 0.078 |

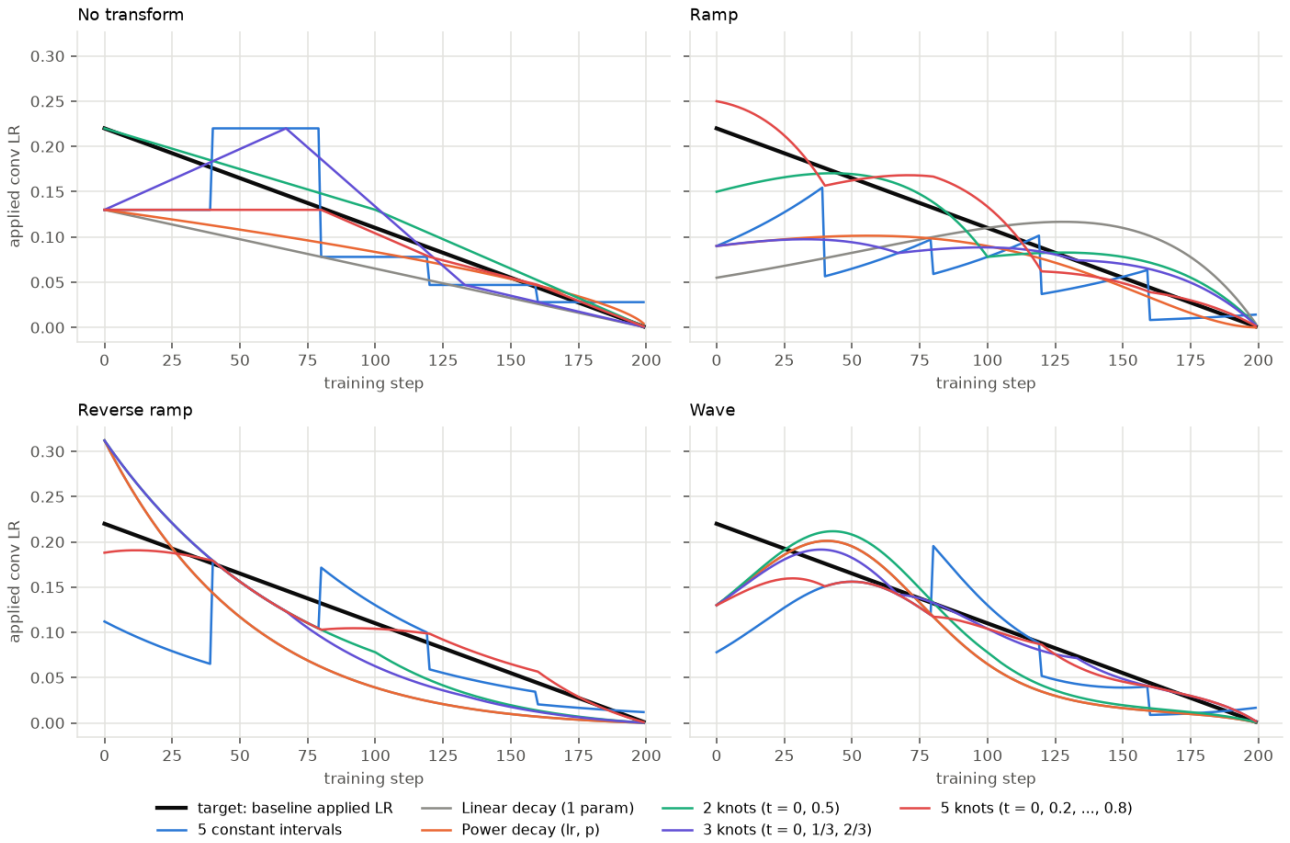



Figure 3. Applied conv LR (spec x transform) of each family's best schedule; the black line is the baseline's applied LR, which a perfect search would reproduce.

### *Observations*

Held-out means over 8 seeds have a standard error of about 0.045 pp, so differences below about 0.1 pp are not resolved. Target: the baseline's 93.92 on the same seeds.

5 knots is the only family that reached the target in all four conditions in these searches (93.91-93.96), and its start, the oracle projected onto the family, is already there (93.91-93.97): 5 knots on the grid represent all four target schedules.

5 constant intervals of conv reach only 93.62 here without a transform (start 93.52), but that is this search's local optimum, not the family's limit. Earlier 5-interval searches at batch size 2000 (same held-out seeds) did better: conv only 93.77 (interval search; conv 0.13, 0.13, 0.078, 0.047, 0.01) and 93.72 (global_interval), and with all four LRs as 5 intervals 93.92 (interval search), matching the baseline, and 93.85 (global_interval). This search kept the last interval at 0.028 where those went to 0.01-0.017: a single-seed coordinate search can stop one grid step short at the end. Under the transforms it reaches 93.65 / 93.86 / 93.82 (ramp / reverse ramp / wave; projected starts 93.32 / 93.81 / 93.11).

Linear decay matches the target without a transform (93.86-93.92) and under wave (93.91), but no initial LR compensates a ramp (93.42) or a reverse ramp (93.59): the shape is wrong, not the scale. Power decay fixes the ramp (93.95 with p = 2, lr 0.36: decaying faster) but not the reverse ramp (93.59): (1-t)^p cannot rise early, which the reverse ramp needs.

2 and 3 knots land in between (93.68-94.06), with no consistent ordering across conditions.

Single-seed search sometimes leaves a good start for a worse held-out schedule (no transform: linear 93.92 -> 93.86, 3 knots 93.94 -> 93.86, 5 constant intervals moved its first value from 0.22 to 0.13): the search optimizes seed 0's test score, the same winner's curse as before.

## Experiment 4. Hparam range from scratch (Experiment 2 redone)

### *Question*

Experiment 2's +-5 sweeps depend on where the center landed. How far can each LR move from the center, on each side, before the score drops by 10%? A range that ends on both sides marks an LR with an optimum; a side that never drops marks an LR the score is insensitive to in that direction.

### *Setup*

| Item | Value |
|---|---|
| Model / score | As in Experiment 2: baseline_bs2000 except the four LRs, constant for the first L steps from initialization; test TTA right after the L steps; 1 seed (0) |
| Center | Experiment 2's joint-search center for each L (re-evaluated; it reproduces Experiment 2's TTA) |
| Probe | One LR at a time (others at the center), one 0.6^k grid step at a time on each side (+1 = x1/0.6, larger), until TTA <= 0.9 x the center's TTA (a 10% relative drop) or 20 steps. A diverged run counts as a drop |

### *Stop offsets (smaller side / larger side) and the LR range within 10% of the center*

| LR | L = 1 | L = 2 | L = 5 | L = 10 | L = 40 |
|---|---|---|---|---|---|
| conv | -2 / +3<br>0.078 - 0.36 | -2 / +3<br>0.13 - 0.6 | -2 / +3<br>0.047 - 0.22 | -2 / +3<br>0.047 - 0.22 | -3 / +3<br>0.028 - 0.22 |
| head | <-20 / +7<br>0.0022 - 1300 | <-20 / +14<br>3.7e-05 - 770 | <-20 / >20<br>2.2e-07 - 170 | <-20 / >20<br>6.1e-07 - 460 | <-20 / +3<br>0.078 - 5900 |
| norm_bias | <-20 / >20<br>2.9e-08 - 21 | <-20 / +17<br>6.1e-05 - 5900 | <-20 / +3<br>0.078 - 5900 | <-20 / +3<br>0.078 - 5900 | <-20 / +6<br>0.01 - 3500 |
| whiten_bias | <-20 / +10<br>0.0022 - 5900 | <-20 / +5<br>0.047 - 9800 | <-20 / +6<br>0.047 - 16000 | <-20 / +17<br>0.00047 - 46000 | <-20 / +11<br>0.078 - 350000 |

<-20 / >20: no 10% drop within 20 steps on that side (the range shown then ends at the last probed point). Center TTA: L = 1: 25.43, L = 2: 36.89, L = 5: 52.00, L = 10: 59.47, L = 40: 84.60.

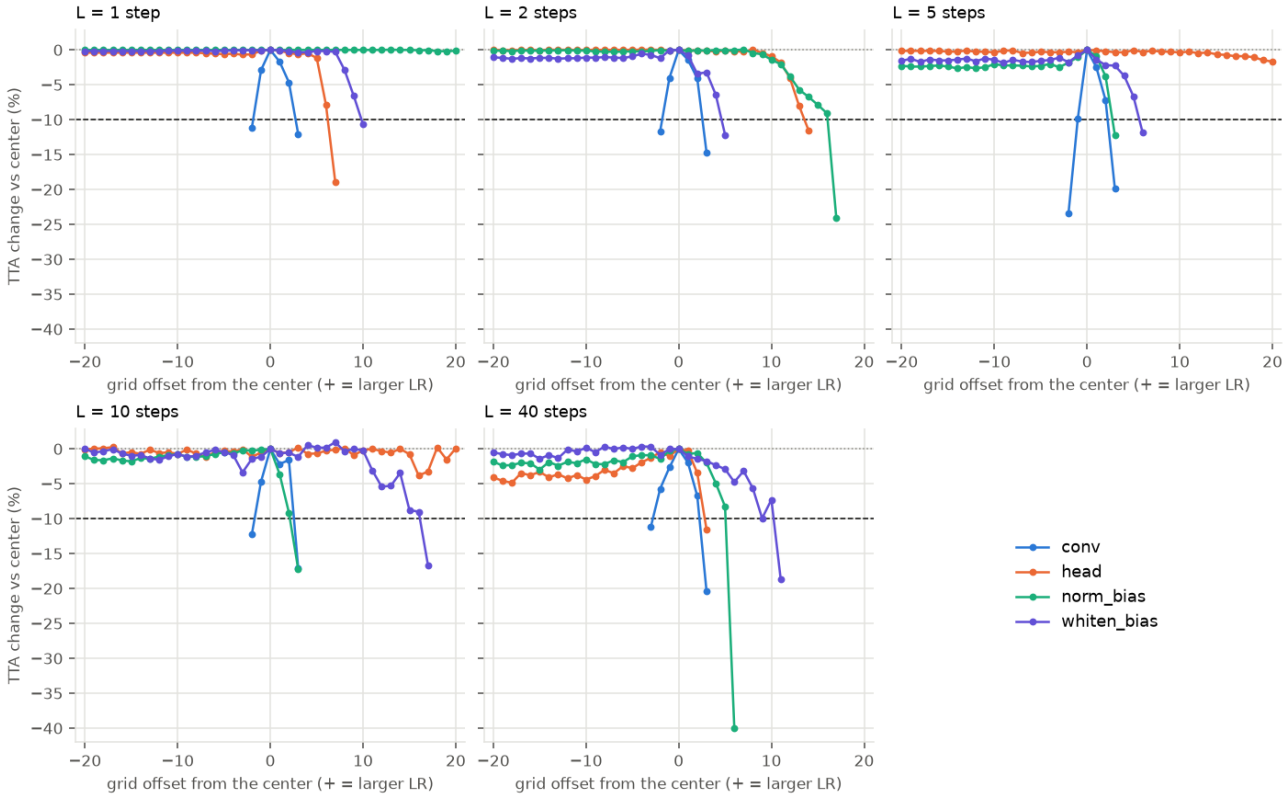



Figure 4. Relative TTA change along each probe; the dashed line is the 10% stop. Points below -40% are drawn at -40%.

### *Observations*

One seed: single-run noise is about 0.13 pp at the full 200 steps; the stop criterion (a 10% relative drop: 2.5 pp at L = 1, 8.5 pp at L = 40) is far above it, so the stop offsets are not noise.

Conv is the only LR with an optimum: its range ends on both sides at every L, at -2 or -3 grid steps (smaller) and +3 (larger). The LR range within 10% of the center spans 5-6 grid points, a factor of 4.6-7.9 in LR (0.078-0.36 at L = 1, 0.028-0.22 at L = 40).

Head, norm_bias and whiten_bias never drop by 10% on the smaller side within 20 steps (down to 0.6^20 = 3.7e-5 x the center) at any L. The worst smaller-side loss is 4.10 pp (head, L = 40); most are below 1.5 pp, and at L = 1-2 below 0.5 pp. These three LRs only have an upper limit.

Their larger sides do end, but late and without a pattern across L: head at +7 / +14 / none / none / +3 (L = 1 / 2 / 5 / 10 / 40), norm_bias at none / +17 / +3 / +3 / +6, whiten_bias at +10 / +5 / +6 / +17 / +11. So their Experiment 2 centers mostly mark where "too large" begins, not an optimum: any value below that edge scores about the same.

At L = 10 a few probe points beat the center (whiten_bias +0.57 pp, head +0.15 pp): the joint-search center is not an exact optimum on one seed, which is the center dependence this experiment was meant to remove.

## Experiment 5. Which non-conv interval schedules help

### *Question*

The two original 5-interval searches scored 93.77 (conv only) and 93.92 (all four LRs) on held-out seeds, but their conv schedules differ in the last two intervals (0.078, 0.017 vs 0.047, 0.01). With conv fixed, do the other groups' searched intervals help, and which group?

### *Setup*

| Item | Value |
|---|---|
| Schedules | C4 = conv constant 0.13, 0.13, 0.078, 0.047, 0.01 (5 x 40 steps; the four-LR interval search's conv). Searched intervals: head 2100, 2100, 2100, 2100, 1300; norm_bias 770, 770, 770, 460, 280; whiten_bias 460, 460, 280, 280, 280. Baseline: conv 0.22 -> 0, head 1300 -> 0, norm_bias 99 -> 0 (linear over 200 steps), whiten_bias 36 -> 0 over 75 steps, then 0 |
| Runs | Plain training runs (no search) on held-out seeds 100-107, current code; the paired difference is per seed against the first row |

### *Held-out TTA*

| Conv | head | norm_bias | whiten_bias | Held-out TTA (mean +- SE) | vs first row (paired) |
|---|---|---|---|---|---|
| C4 | baseline | baseline | baseline | 93.95 +- 0.06 | - |
| C4 | searched | baseline | baseline | 93.90 +- 0.05 | -0.05 +- 0.08 |
| C4 | baseline | searched | baseline | 93.77 +- 0.04 | -0.18 +- 0.08 |
| C4 | baseline | baseline | searched | 93.88 +- 0.05 | -0.07 +- 0.07 |
| C4 | searched | searched | searched | 93.92 +- 0.04 | -0.03 +- 0.07 |
| baseline (0.22 -> 0) | searched | searched | searched | 93.87 +- 0.04 | -0.08 +- 0.07 |
| baseline (0.22 -> 0) | baseline | baseline | baseline | 93.92 +- 0.05 | -0.03 +- 0.07 |

For reference, the original conv-only interval search's schedule (conv 0.13, 0.13, 0.078, 0.078, 0.017, the others on the baseline) scored 93.77 on the same seeds (older code).

### *Observations*

The other groups' searched intervals do not help. With conv fixed at C4 and the other three groups on the baseline, the held-out TTA is 93.95 +- 0.06, as high as any row. Switching all three to their searched intervals gives 93.92 (-0.03 +- 0.07 paired); switching one at a time gives head -0.05 +- 0.08, whiten_bias -0.07 +- 0.07 and norm_bias -0.18 +- 0.08 (about 2.3 SE: norm_bias's searched intervals, 770 -> 280 instead of 99 -> 0, probably hurt).

The 93.77 vs 93.92 gap between the original conv-only and four-LR searches came from conv, not from the other groups. Both conv schedules with the other groups on the baseline: 0.13, 0.13, 0.078, 0.078, 0.017 scores 93.77 and C4 (0.13, 0.13, 0.078, 0.047, 0.01) 93.95, +0.18 +- 0.05 paired (7 of 8 seeds). The four-LR rerun reproduces the original run's per-seed scores exactly, so the older 93.77 run is directly comparable.

So 5 constant intervals of conv alone do match the baseline (93.95 vs the baseline's 93.92 on these seeds) when the last two intervals are low enough (0.047, 0.01); Experiment 3's conv-only search stopped at 0.047, 0.028.

The searched non-conv intervals with the baseline's linear conv score 93.87 (-0.08 +- 0.07 vs the C4 control), close to the baseline itself: the other groups' schedules matter little with either conv schedule.

## Experiment 6. Seed noise of the baseline

### *Question*

How much does the baseline's score vary across training seeds, and how many seeds does a comparison need?

### *Setup*

| Item | Value |
|---|---|
| Runs | baseline_bs2000, plain training runs on seeds 0-127 (seed 0 is the search seed; 100-107 are the held-out seeds); the seed sets the initialization, data order and augmentation |
| Score | Test TTA accuracy (%) and test accuracy without TTA |

### *Baseline across seeds*

| Metric | Seeds | Mean | SD (one run) | SE of the mean | Min | Max |
|---|---|---|---|---|---|---|
| TTA | 128 | 93.942 | 0.121 | 0.011 | 93.61 | 94.28 |
| No TTA | 128 | 93.164 | 0.140 | 0.012 | 92.69 | 93.52 |

### *Averaging over seeds*

| Seeds per mean (n) | SD of an n-seed mean: SD/sqrt(n) | Measured (disjoint blocks of n) | Smallest difference two n-seed means resolve (95%) |
|---|---|---|---|
| 1 | 0.121 | 0.121 (128 blocks) | 0.34 pp |
| 2 | 0.086 | 0.089 (64 blocks) | 0.24 pp |
| 4 | 0.061 | 0.069 (32 blocks) | 0.17 pp |
| 8 | 0.043 | 0.051 (16 blocks) | 0.12 pp |
| 16 | 0.030 | 0.041 (8 blocks) | 0.08 pp |
| 32 | 0.021 | 0.028 (4 blocks) | 0.06 pp |

Seed 0 (the search seed): 93.92. Held-out seeds 100-107: mean 93.92. All seeds: 93.94.

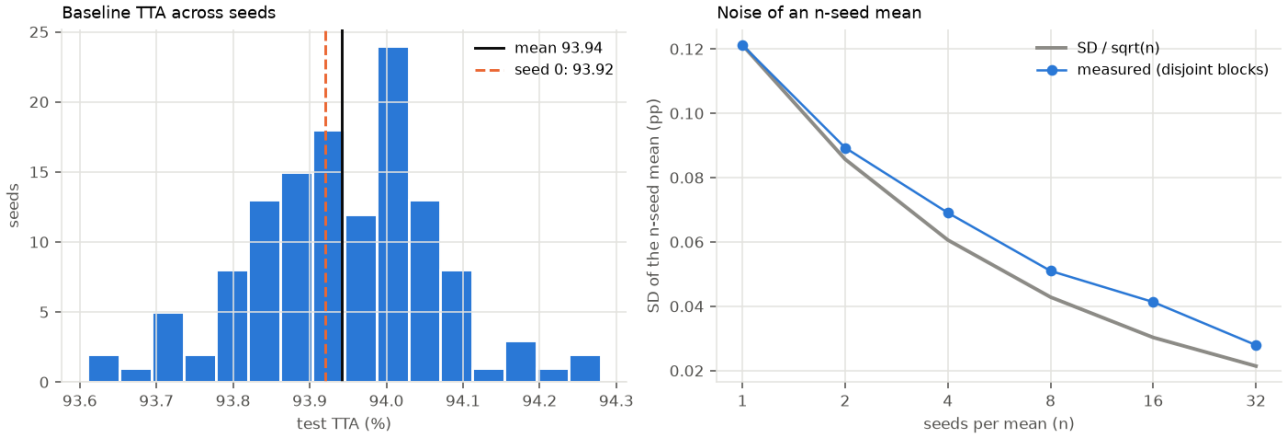



Figure 5. Left: the baseline's TTA over seeds. Right: the spread of an n-seed mean, predicted from the single-run SD and measured on disjoint blocks of seeds.

### *Observations*

One run of the baseline has a seed SD of 0.121 pp in TTA (128 seeds: mean 93.94, range 93.61-94.28, 0.67 pp from worst to best seed); without TTA the SD is 0.140 pp (mean 93.16). The mean over all 128 seeds is known to +-0.011 pp (SE).

Averaging over seeds shrinks the noise about as SD / sqrt(n): measured on disjoint blocks of seeds, an 8-seed mean has SD 0.051 (predicted 0.043; 16 blocks, so the measured value itself is uncertain by about +-20%). Telling two schedules apart at 95% takes a difference of about 0.34 pp with 1 seed each, 0.12 pp with 8 seeds and 0.06 pp with 32 seeds (unpaired; pairing by seed helps only if the seeds' effects are shared).

The seeds used so far are typical: seed 0 (the search seed) scores 93.92 and the held-out seeds 100-107 average 93.92, both within 0.02 pp of the 128-seed mean.

## Experiment 7. Hparam range around the baseline

### *Question*

Starting from the baseline instead of a from-scratch center, how far can each LR's initial value move, on each side, before the full run's score drops by 10%?

### *Setup*

| Item | Value |
|---|---|
| Runs | baseline_bs2000, full 200-step runs, seed 0. One LR at a time (the others on the baseline) |
| Probe | The LR's whole baseline schedule is scaled so its initial value steps along the 0.6^k grid (+1 = x1/0.6, larger), on each side until TTA <= 0.9 x the baseline's TTA on seed 0, or 20 steps. A diverged run counts as a drop |

### *Stop offsets (baseline TTA on seed 0: 93.92; stop at 84.53)*

| LR | Baseline initial LR | Stop (smaller / larger) | Initial LR range within 10% |
|---|---|---|---|
| conv | 0.22 | -6 / +6 | 0.017 - 2.8 (x165) |
| head | 1300 | <-20 / +8 | 0.047 - 46000 (x9.79e+05) |
| norm_bias | 99 | <-20 / +9 | 0.0036 - 5900 (x1.64e+06) |
| whiten_bias | 36 | <-20 / >20 | 0.0013 - 980000 (x7.54e+08) |

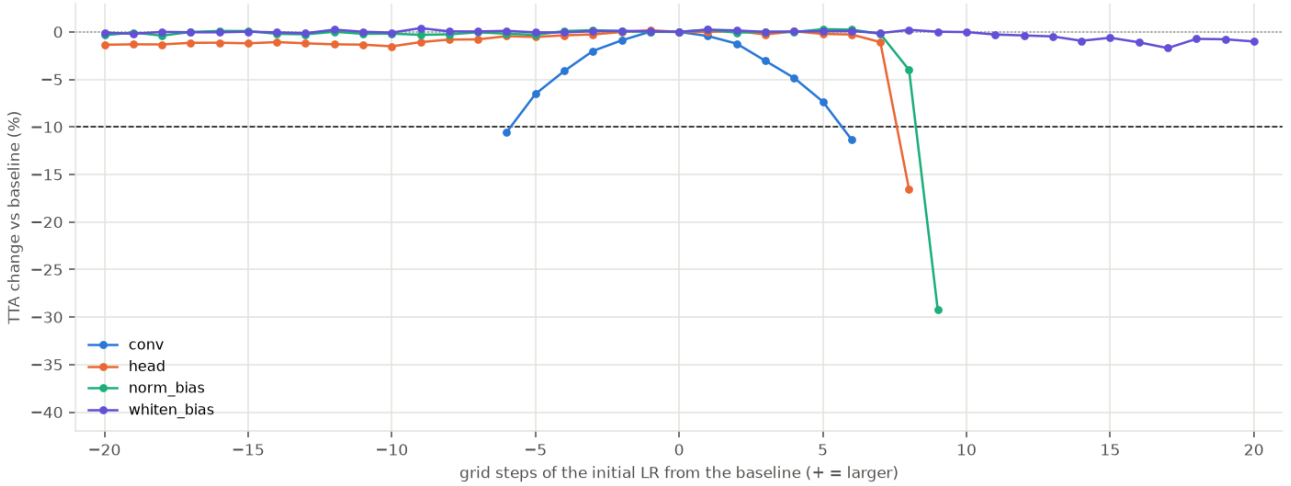



Figure 6. Relative TTA change of full runs as one LR's initial value moves from the baseline; dashed: the 10% stop. Points below -40% are drawn at -40%.

### *95% confidence set for the best initial LR*

Each probed value is one run on seed 0, so its score carries the single-run noise of Experiment 6 (SD 0.121 pp, assumed the same at every value and independent between values). A value is kept when its score is within c of the best observed score, with c = q x SD and q the 95% quantile of max_j (Z_j - Z_1) over the k probed values (Z independent standard normals; Monte Carlo). This is Gupta's subset selection: the set contains the best probed value with probability at least 95%, even in the hardest case where all values are equally good. (open) marks an end that reaches the probe limit on a side without a 10% drop, so the set may continue beyond it.

| LR | Probed points k | Cutoff c (pp) | Best observed | 95% set (initial LR) | Points in set |
|---|---|---|---|---|---|
| conv | 13 | 0.43 | 0.13 (93.95) | 0.13 - 0.22 | 2 of 13 |
| head | 29 | 0.47 | 770 (94.08) | 280 - 27000 | 10 of 29 |
| norm_bias | 30 | 0.47 | 1300 (94.19) | (open) 0.006 - 3500 | 22 of 30, with gaps |
| whiten_bias | 41 | 0.48 | 0.36 (94.29) | (open) 0.0013 - 5900 | 28 of 41, with gaps |

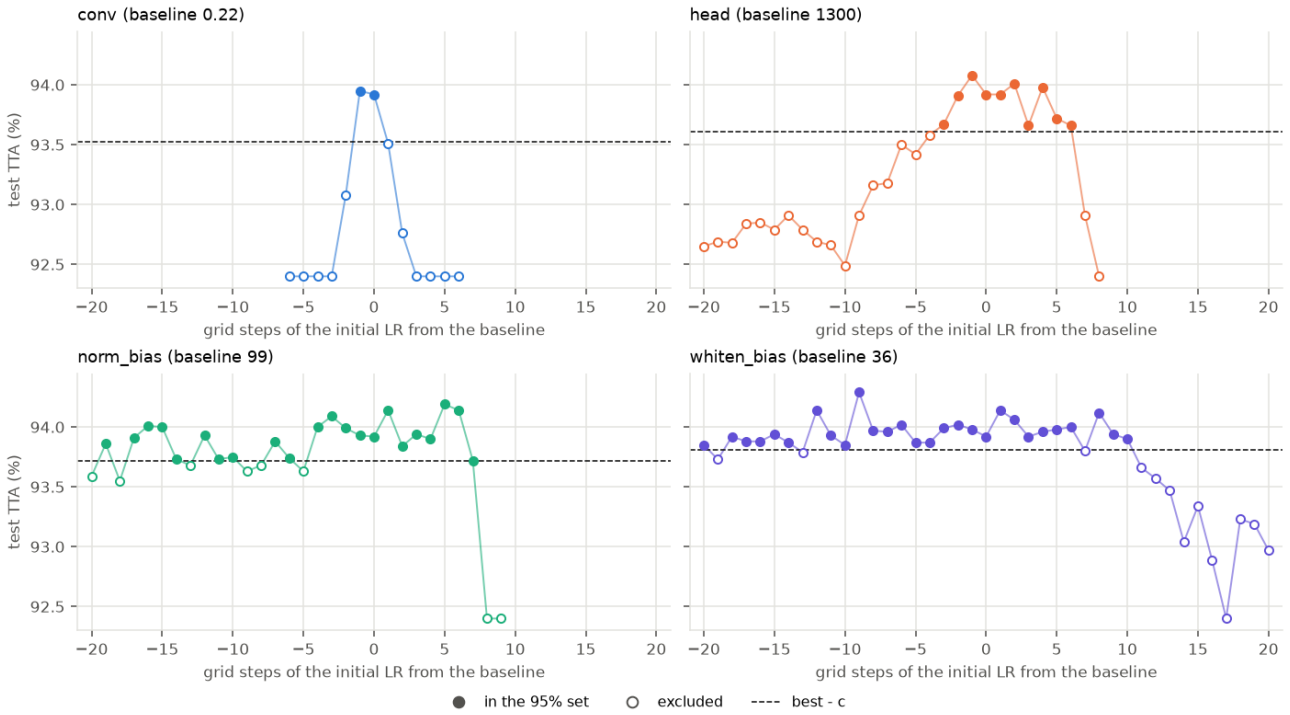



Figure 7. The probes near the top: filled points are in the 95% set (score at least best - c, dashed); points below 92.4 are drawn at 92.4.

### *Observations*

One seed (0); the 10% stop (84.53) is about 80 single-run SDs (Experiment 6) below the baseline, so the stop offsets are not noise; differences under about 0.34 pp between single points are.

Conv has a clear optimum, and the baseline sits on it: 0.13 scores 93.95 and 0.22 (baseline) 93.92; one step further costs 0.4-0.8 pp (0.36: 93.51; 0.078: 93.08). Its 10% range ends at -6 (0.01: 84.03) and +6 (4.6: 83.28), so in full runs conv tolerates a factor of about 165 (0.017 to 2.8) before losing 10%, much wider than in the from-scratch L-step setting of Experiment 4 (5-6 grid points).

Head, norm_bias and whiten_bias never drop by 10% on the smaller side within 20 steps (down to 3.7e-5 x the baseline). Norm_bias and whiten_bias are flat there (all within 93.55-94.29, within about 0.4 pp of the baseline, roughly the single-run noise range); head loses about 1.2 pp once it is below about 8 (92.5-92.9 from 7.7 down to 0.047), still far from 10%.

On the larger side, head stops at +8 (76000: 78.35; 46000 still 92.91) and norm_bias at +9 (9800: 66.44; 5900: 90.16), both after 6-7 steps of no loss: they fail by blowing up, not gradually. whiten_bias never stops: up to 980000 (x27000) it stays above 92.3.

So around the baseline only conv needs tuning; the three other LRs only need to stay below a blow-up edge about 35x above their baseline values (head, norm_bias; the last points without loss are 46000 and 3500), and whiten_bias barely matters at all.

95% confidence sets for the best initial LR (single runs with SD 0.121 pp; cutoff c = 0.43-0.48 pp below the best observed score, wider for more probed points). Conv: 0.13-0.22, the two values at the top (93.95, 93.92); the next ones are excluded (0.36: 93.51, only 0.01 pp under the cutoff 93.52; 0.078: 93.08), so the best conv initial LR is pinned to two grid points, and the baseline's 0.22 is one of them.

Head: 280-27000 (10 of 29 points, contiguous), a factor of about 100 around the best observed 770; below 280 the scores fall off (170: 93.58), and 46000 is 1.0 pp below the baseline (92.91). The baseline's 1300 is inside.

Norm_bias: anything from the smallest probed values (0.006; open, as the probe never dropped 10% on that side) up to 3500, with gaps: the scores below 3500 are within noise of each other, so the data do not locate a best value, only an upper edge. Whiten_bias: likewise from 0.0013 (open) up to 5900; from 9800 upward the scores fall (93.66 and lower) and are excluded.